Supplementary Material
Beautified Is Not Forged: Measuring and Separating Beautification from Face Forgery

S1 Pre-registration and what was read when

Every bar quoted in the paper was written into a dated file before the corresponding model was trained or the corresponding test source was read. Celeb-DF-B and the DFD filter set were read once per model, after the checkpoint had been chosen on validation data. The Celeb-DF-v2 filter set was used to diagnose which filters cause the damage ("Where the damage comes from" in the paper), so it is a development source for every design choice made after that point; the DFD filter set was built afterwards from a second filter library. No per-filter result of the DFD set was inspected. The checkpoint of every model is chosen on validation data only: FF++ validation macro-F1 (three-way, FF++ only), FF++ validation AUC (binary), or the mean of FF++ validation macro-F1 and balanced accuracy on held-out originals and renders of the two training services (three-way, full data). An earlier selection rule (F1 of the filter class on an imbalanced validation set) picked the epoch with the lowest specificity on untouched originals; it was replaced before the models of this paper were trained, and the earlier model appears only in Table S17.

The arms of the smoothing chain (Sec. S9) were declared one at a time, each after the previous result had been read and before its own training: the bars and the allowed readings are quoted with each result. The DF40 evaluation list, detectors and statistics were fixed before any detector was run on it; the scoring of the exposure arms and of CADDM on DF40 was added afterwards and is marked post hoc. The fusion pairs, rules and reading were fixed before any fused score was computed.

S2 Cases near the class boundary

Table S1: Cases near the class boundary and whether they occur in our data.

Case	Forged?	In our data
Camera processing, JPEG, video coding	no	all sets
Make-up worn at capture	no	incidental
Beautification by an app or a service, neural or not	no (beautified)	scripted, 3 services, app presets
Reenactment by another person's performance	yes	Face2Face, NeuralTextures
A part from another face, or inpainted	yes	part-level forgeries
Face swap	yes	FF++, Celeb-DF, DFD
Beautified forgery	yes (and beautified)	FF++, Celeb-DF-B
Expression editing; synthesised face	yes	not tested

S3 Each baseline under its own preprocessing

Table S2: Reported and reproduced. Published numbers are from DeepfakeBench (frame level, eight CNNs) and from the Effort and Forensics Adapter papers. "DFB crop": our run of the released weights on DeepfakeBench crops (dlib frontal detector, 81 landmarks, five-point similarity alignment, scale 1.3, 256 px, 32 frames per video by linear spacing), built with the benchmark's own alignment function. "Our crop": landmark box with a 35% margin, aspect kept. Videos common to both frame sets (516 for Celeb-DF-v2, 332 for DFD). Our DFD set is a 332-video subset, so no published DFD number is comparable and none is listed as a target.

	Celeb-DF-v2 frame AUC		Celeb-DF-v2 video AUC		DFD frame AUC	
Detector	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop
Xception	0.737	0.744 / 0.718	–	0.821 / 0.772	–	0.845 / 0.854
EfficientNet-B4	0.749	0.754 / 0.750	–	0.813 / 0.807	–	0.818 / 0.810
F3Net	0.735	0.732 / 0.708	–	0.793 / 0.750	–	0.781 / 0.781
SPSL	0.765	0.732 / 0.703	–	0.803 / 0.752	–	0.812 / 0.829
UCF	0.753	0.775 / 0.744	–	0.841 / 0.788	–	0.818 / 0.822
RECCE	0.732	0.744 / 0.688	–	0.827 / 0.731	–	0.839 / 0.833
CORE	0.743	0.744 / 0.715	–	0.814 / 0.764	–	0.830 / 0.827
SRM	0.755	0.763 / 0.717	–	0.844 / 0.778	–	0.837 / 0.830
Effort [1]	0.901	0.879 / 0.881	0.956	0.927 / 0.930	0.923	0.933 / 0.939
Forensics Adapter [2]	0.900	0.899 / 0.887	0.957	0.956 / 0.945	0.933	0.940 / 0.940
GenD [3]	–	0.900 / 0.891	0.960	0.960 / 0.954	–	0.932 / 0.934
DFD-HR [4]	0.910	0.886 / 0.879	0.960	0.951 / 0.943	0.953	0.942 / 0.943

Forensics Adapter reproduces its published Celeb-DF-v2 numbers on its own crop. Effort does not; an independent re-run of its released weights reports 0.932 at video level and a retraining 0.871, against 0.956 in the paper, and our 0.927 is in line with the re-run. The training code released with both detectors keeps the checkpoint with the best test-set AUC; the papers do not state a selection rule. The untreated comparisons on Celeb-DF-v2 and DFD in the paper pair our models on our crop with each baseline on its own crop; on the retouching service and on DF40 all detectors share one crop.

CADDM and LAA-Net are run with the authors' code, released weights and test-time preprocessing. CADDM's released EfficientNet-B4 checkpoint gives a Celeb-DF-v2 video AUC of 0.937 on the authors' own crop against 0.939 published; in our pipeline it re-aligns our crop with its own landmark model. LAA-Net's released SBI checkpoint gives 0.943 against 0.954 published on its own crop, which we rebuild inside our face box with a different face detector from the authors'; the checkpoint was trained on uncompressed FF++ frames. On their own crops Forensics Adapter reaches a Celeb-DF-v2 frame AUC of 0.899 and Effort 0.879 (published 0.900 and 0.901), and the eight CNNs gain 0.00–0.05 over our crop. At video level Ours is at 0.947 on Celeb-DF-v2 and 0.970 on DFD. The binary control (0.914 and 0.967 at frame level) is above Effort with intervals excluding zero and on par with Forensics Adapter; a model trained on FF++ alone with our synthesised part-level forgeries does not reach it, so the margin of the full-data models comes from the external real data (FFHQ stills and commercial renders), not from the synthesis.

S4 Where the filter is applied

Table S3: The same videos, frames and per-video filter assignment with the filter applied to the face crop or to the full frame before the crop. A filter with a vignette is centred on the face in the first case. The paper reports the full-frame version.

	Celeb-DF-v2, 11 filters			DFD, 16 filters		
Detector	AUC	AUC_f crop	AUC_f full frame	AUC	AUC_f crop	AUC_f full frame
Xception	0.718	0.673	0.693	0.854	0.823	0.834
EfficientNet-B4	0.750	0.698	0.725	0.810	0.777	0.781
F3Net	0.708	0.663	0.688	0.781	0.769	0.774
SPSL	0.703	0.664	0.683	0.829	0.834	0.837
UCF	0.744	0.669	0.697	0.822	0.793	0.804
RECCE	0.688	0.636	0.661	0.833	0.795	0.808
CORE	0.715	0.669	0.693	0.827	0.813	0.821
SRM	0.717	0.696	0.712	0.830	0.811	0.822
SBI [5]	0.817	0.729	0.776	0.829	0.803	0.810
UnivFD	0.662	0.610	0.622	0.840	0.793	0.801
NPR	0.522	0.502	0.497	0.473	0.541	0.516
Effort [1]	0.881	0.788	0.813	0.939	0.904	0.905
Forensics Adapter [2]	0.887	0.790	0.814	0.940	0.907	0.924
GenD [3]	0.891	0.818	0.856	0.934	0.913	0.925
DFD-HR [4]	0.879	0.781	0.824	0.943	0.921	0.928
OURS	0.891	0.849	0.873	0.942	0.927	0.931
OURS, FF++ only	0.898	0.797	0.851	0.943	0.925	0.932
binary control	0.914	0.863	0.883	0.961	0.949	0.952

S5 The decomposition of a commercial render

Algorithm 1 is the procedure of the Method section of the paper: dense optical flow on locally normalised luminance gives the warp f between an original O and its render R, the photometric residual is taken in the geometry of O, and four single-operation components are formed, each with its measured strength and exact footprint.

Algorithm 1 Single-operation components of a commercial render

Require: original O, render R, eye mask E, σ, noise floors τ_k

$f \leftarrow \text{DIS}\big(\text{LCN}(O), \text{LCN}(R)\big)$

$P \leftarrow R \circ (\text{id} - f^{-1}) - O$

$C_{\text{eye}} \leftarrow \mathcal{W}(O; E \odot f)$; $\quad C_{\text{contour}} \leftarrow \mathcal{W}(O; (1-E) \odot f)$

$C_{\text{tone}} \leftarrow O + G_\sigma * P$; $\quad C_{\text{texture}} \leftarrow O + P - G_\sigma * P$

for $k \in \{\text{eye}, \text{contour}, \text{tone}, \text{texture}\}$ **do**

  $q_k \leftarrow \text{Measure}_k(C_k, O)$; **if** $q_k \geq \tau_k$ **keep** $(C_k,\ e_k,\ M(C_k))$

end for

S6 Exposure arms on every source

Table S4: Which class receives beautified training views. FF++ only, forged head only, everything else identical; three seeds. Celeb-DF-B under the one rule of the paper.

Arm	Beautified training views	AUC_b (3 seeds)	$\Delta\text{MISS}_{\text{beauty}}$ (3 seeds)	FA_b
P0	no beautified views	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	5.1 / 5.6 / 6.8
P1	beautified genuine faces only	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	4.5 / 3.9 / 3.7
P2	beautified genuine and forged faces	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	5.6 / 5.3 / 5.4
Paired differences, 95% video-cluster intervals (seeds 1 / 2 / 3)				
P1 − P0	$\Delta\text{MISS}_{\text{beauty}}$	+4.5 [1.5, 7.6] / +7.5 [4.4, 10.8] / +9.5 [6.7, 12.3]		
P2 − P0	$\Delta\text{MISS}_{\text{beauty}}$	-7.2 [-9.4, -5.0] / -4.0 [-6.1, -2.0] / +2.0 [0.1, 4.0]		
P2 − P1	$\Delta\text{MISS}_{\text{beauty}}$	-11.7 [-14.5, -8.8] / -11.5 [-14.5, -8.9] / -7.5 [-10.0, -5.0]		

Table S5: Which class receives beautified training views (P0 none, P1 genuine only, P2 both), FF++ only, forged head only; values per seed. Effort scheme: the SVD-residual tuning of the released Effort code in place of LoRA. Filters on the full frame; ΔMISS at the 5%-FPR threshold of untreated genuine frames.

		Celeb-DF-B			Celeb-DF-v2 filters		DFD filters	
Tuning	Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	AUC_f	ΔMISS	AUC_f	ΔMISS
LoRA	P0	0.927 / 0.937 / 0.944	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.852 / 0.858 / 0.855	+7.1 / +4.4 / +8.3	0.924 / 0.928 / 0.932	+9.3 / +7.2 / +8.7
LoRA	P1	0.928 / 0.931 / 0.939	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.798 / 0.820 / 0.838	+16.6 / +17.7 / +16.3	0.905 / 0.911 / 0.913	+11.2 / +9.3 / +9.2
LoRA	P2	0.929 / 0.933 / 0.935	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.848 / 0.850 / 0.856	+0.3 / +3.0 / -0.9	0.929 / 0.929 / 0.927	+2.5 / +1.8 / -0.5
Effort scheme	P0	0.943 / 0.948	0.882 / 0.886	+10.1 / +10.7	0.876 / 0.886	+2.3 / -1.3	0.942 / 0.947	+8.7 / +7.4
Effort scheme	P2	0.938 / 0.943	0.889 / 0.890	+5.5 / +5.5	0.872 / 0.879	-7.6 / -8.4	0.948 / 0.953	+3.0 / +2.4

Paired video-cluster intervals for P2 minus P0 on the two filter sources: the miss rise falls by 5.0 to 9.9 points in nine of ten seed-by-source cells (intervals below zero; the tenth is -1.4 $[-3.3, 0.6]$), the false-alarm rise grows by 2.0 to 4.6 points in the same nine, and the AUC under the filters changes by -0.008 to $+0.006$ (intervals contain zero except for the Effort scheme on DFD, $+0.006$). For P1 minus P0 on Celeb-DF-v2 the miss rise grows by 13.3 and 8.0 points and the AUC under the filters falls by 0.038 and 0.017.

S7 The feature-space probe, per filter

Table S6: FF++-only CLIP model, FF++ validation frames under the eleven held-out filters: cosine of the mean feature shift with the forged weight vector w, mean change of the forged logit on genuine and forged frames, and share of the shift energy inside the 16-dimensional subspace spanned by training-time beautifications.

Held-out filter	n	$\cos(w, \bar{\Delta})$	Δlogit genuine	Δlogit forged	energy in V_{16}
earlybird	88	0.20	+7.84	+1.10	0.86
mayfair	94	0.33	+5.64	+1.80	0.83
kelvin	92	0.04	+2.40	-0.71	0.90
clarendon	99	0.07	+1.90	+0.40	0.86
lofi	95	0.06	+1.86	+0.17	0.87
moon	89	0.06	+1.75	-0.10	0.75
perpetua	107	0.09	+0.99	+0.21	0.77
inkwell	95	0.09	+0.86	-0.25	0.63
reyes	104	0.02	+0.76	-0.27	0.89
slumber	101	0.00	+0.43	-0.28	0.87
brooklyn	95	0.00	+0.14	-0.04	0.88

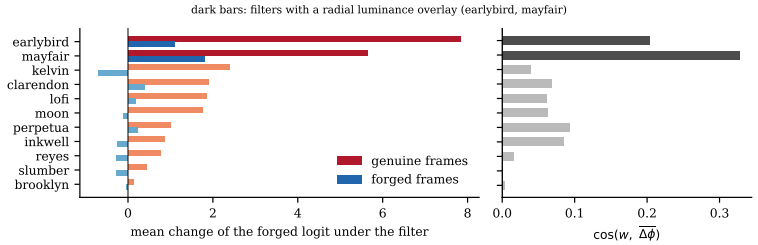


Figure S1: Mean change of the forged logit under each held-out filter (left) and cosine of the mean feature shift with the forged weight vector (right), FF++-only CLIP model on FF++ validation frames. Dark bars: the two filters with a radial luminance overlay.

S8 Linear erasure of the beautification subspace

Table S7: FF++-only CLIP model, FF++ validation frames under the eleven held-out filters. The feature shifts caused by training-time beautifications are decomposed by SVD on training frames, the first k directions are projected out, and a logistic forged head is refitted on training features. FA and MISS at the 5%-FPR threshold of clean genuine frames.

Forged head	AUC clean	AUC filtered	FA filtered	MISS filtered
trained head	0.987	0.953	21.5	5.9
refit, 0 directions erased	0.985	0.916	15.8	16.6
refit, 2 directions erased	0.933	0.907	18.5	15.1
refit, 8 directions erased	0.932	0.844	18.3	27.6
refit, 32 directions erased	0.919	0.804	19.2	32.1

The two leading directions hold 79% of the shift energy of training-time beautifications. Removing them costs 0.05 clean AUC and does not lower the false alarms under the held-out filters, so the directions that beautification moves along carry forgery evidence.

S9 Skin smoothing: the chain of pre-registered arms

This section gives the full record behind the section of the paper on skin smoothing. All arms are FF++-only models with the forged head only; they share the architecture, data, steps and checkpoint rule of the exposure arms and differ only in which class receives beautified training views and in what those views contain. No image of any retouching service is used in any arm. The scripted smoothing in the views is a bilateral filter blended inside a feathered face hull, the same kind of edit as the service's (face-local, edge-preserving: high-frequency energy on flat skin 0.68 at the top of the training range against 0.60 for the service's level 90, background 1.00 against 0.96), and its strength range was set to cover the levels measured on two other services whose renders are in RetouchingFFHQ; the test service's measurements existed when that range was set. What is tested is robustness across implementations and image domains of one operation at matched strength, not generalisation to an edit the views never contained.

Table S8: Commercial skin smoothing, every detector and arm: 2,210 FFHQ originals and their renders by a service never seen in training. Smoothing-90: the strongest of three levels; share of renders scored above their own original (50 means no signal), medians of $P(\text{fake})$, and share above the threshold that accuses 5% of the originals. All renders: the four operations at three levels (26,519 images). NPR scores originals and renders at 1.0. P0, P1, P2: beautified training views on no class, on genuine faces only, on both classes.

	Smoothing-90 renders			all renders	
Detector	above own original (%)	median $P(\text{fake})$ orig. $\rightarrow$ render	above 5%-FPR thr. (%)	above 5%-FPR thr. (%)	AUROC render vs orig.
Xception	95.2	0.291 $\rightarrow$ 0.944	51.6	12.2	0.563
EfficientNet-B4	93.7	0.316 $\rightarrow$ 0.964	36.3	9.4	0.545
F3Net	97.8	0.131 $\rightarrow$ 0.974	50.6	11.8	0.568
SPSL	97.3	0.463 $\rightarrow$ 0.963	51.2	12.1	0.567
UCF	95.4	0.031 $\rightarrow$ 0.984	41.4	11.2	0.575
RECCE	96.5	0.120 $\rightarrow$ 0.944	36.8	9.5	0.558
CORE	96.6	0.249 $\rightarrow$ 0.947	54.8	12.4	0.569
SRM	99.3	0.461 $\rightarrow$ 0.668	62.8	13.9	0.570
SBI [5]	96.2	0.208 $\rightarrow$ 0.991	37.3	11.3	0.582
UnivFD	90.4	0.005 $\rightarrow$ 0.019	16.6	7.3	0.526
NPR	n/a (saturated)	n/a	n/a	n/a	n/a
Effort [1]	95.6	0.319 $\rightarrow$ 0.595	43.2	14.8	0.599
Forensics Adapter [2]	93.3	0.058 $\rightarrow$ 0.354	39.7	10.2	0.541
GenD [3]	98.5	0.228 $\rightarrow$ 0.921	68.7	16.8	0.596
DFD-HR [4]	99.0	0.210 $\rightarrow$ 0.711	64.9	15.7	0.554
CADDM [6]	60.9	0.538 $\rightarrow$ 0.745	6.3	5.6	0.523
LAA-Net [7] (own crop rebuilt)	89.2	0.234 $\rightarrow$ 0.969	17.2	7.6	0.578
OURS, three-way, full data	32.5	0.002 $\rightarrow$ 0.001	8.6	4.7	0.371
OURS, symmetric-view variant	30.0	0.003 $\rightarrow$ 0.001	8.1	4.8	0.375
OURS, binary control	70.5	0.000 $\rightarrow$ 0.001	9.1	5.2	0.490
OURS, FF++ only, exposure arms (seed 1)					
P0	98.5	0.049 $\rightarrow$ 0.968	54.2	13.5	0.576
P2	84.1	0.009 $\rightarrow$ 0.116	11.7	5.9	0.528
P2 no smoothing	98.6	0.017 $\rightarrow$ 0.988	55.3	12.7	0.553
P2 smoothing only	27.3	0.053 $\rightarrow$ 0.020	2.3	3.7	0.449
P1	43.5	0.004 $\rightarrow$ 0.004	3.7	5.3	0.500
P1 smoothing only	2.6	0.074 $\rightarrow$ 0.000	0.7	4.9	0.437
P2 symmetric smoothing	53.7	0.017 $\rightarrow$ 0.022	5.9	4.9	0.488
OURS pipeline, FF++ only, Effort's tuning scheme (seed 1)					
Effort scheme, no beautified views	99.1	0.028 $\rightarrow$ 0.963	56.9	13.3	0.587
Effort scheme, symmetric smoothing	88.4	0.029 $\rightarrow$ 0.095	14.5	7.2	0.541
released Effort fine-tuned, no beautified views	98.5	0.091 $\rightarrow$ 0.306	55.3	14.4	0.564
released Effort fine-tuned, symmetric smoothing	96.6	0.080 $\rightarrow$ 0.174	39.9	11.5	0.553

Table S9: Smoothed forgeries and smoothed genuine frames. Celeb-DF-v2 test frames (every fourth of the 32 per video), untreated and under a whole-crop bilateral filter at three strengths, a 5×5 median filter and a 50% blend with a Gaussian blur; the last two forms were shown to no arm. Threshold at 5% FPR on untreated genuine frames. Miss rise: change of the miss rate on forgeries after smoothing, with 95% video-cluster intervals. Last column: frame AUC with both classes under the strong bilateral filter.

		miss rise (points, 95% CI)					false alarms, smoothed genuine (%)					
Arm	miss untr.	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	AUC both smoothed (strong)
P0	42.2	+27.0 [+24.4, +29.7]	+5.1 [-0.3, +10.5]	-16.0 [-21.5, -10.2]	+19.2 [+17.1, +21.2]	+3.0 [+1.4, +5.1]	0.8	16.8	39.6	1.5	5.5	0.747
seed 2	37.7	+26.7 [+24.1, +29.9]	+6.1 [+2.2, +11.8]	-11.8 [-15.5, -5.4]	+18.0 [+16.0, +20.4]	+8.1 [+6.6, +9.9]	1.6	18.2	42.9	1.9	4.4	0.736
seed 3	36.7	+24.0 [+21.4, +26.9]	+14.3 [+8.8, +17.9]	+4.3 [-2.4, +11.0]	+12.6 [+10.6, +14.6]	+5.6 [+4.2, +6.9]	2.3	11.3	25.2	3.0	4.9	0.736
P2	46.8	+16.5 [+14.2, +19.4]	-17.4 [-20.7, -13.1]	-26.5 [-29.7, -21.9]	+13.2 [+11.7, +15.8]	+7.0 [+6.0, +8.9]	3.0	31.8	49.1	2.5	3.3	0.740
P2 no smoothing	46.7	+19.8 [+16.8, +22.0]	-12.2 [-16.3, -6.5]	-28.9 [-32.8, -24.9]	+8.1 [+6.4, +9.8]	+2.2 [+0.7, +3.5]	2.3	25.6	54.4	3.9	4.7	0.732
P2 smoothing only	49.5	-34.3 [-37.6, -30.4]	-47.3 [-51.6, -40.2]	-48.2 [-53.1, -40.8]	+17.2 [+14.7, +20.2]	+10.9 [+9.2, +13.6]	47.8	82.3	88.3	1.7	2.4	0.785
P1	48.8	+45.9 [+39.7, +51.1]	+51.1 [+43.6, +57.2]	+51.2 [+43.7, +57.5]	+29.3 [+25.1, +32.4]	+6.4 [+4.8, +7.6]	0.0	0.0	0.0	0.7	4.7	0.587
P1 smoothing only	43.6	+56.4 [+49.9, +63.7]	+56.4 [+50.0, +63.9]	+56.4 [+50.0, +63.9]	+40.0 [+35.8, +44.4]	+10.8 [+9.1, +12.6]	0.0	0.0	0.0	0.4	4.0	0.643
seed 2	40.0	+58.7 [+52.3, +65.1]	+60.0 [+53.7, +66.7]	+60.0 [+53.7, +66.7]	+38.6 [+34.6, +42.6]	+13.9 [+12.3, +16.1]	0.0	0.0	0.0	0.7	3.1	0.579
seed 3	37.1	+59.3 [+53.9, +64.5]	+62.8 [+57.4, +68.6]	+62.9 [+57.5, +68.7]	+38.0 [+33.8, +41.7]	+13.5 [+11.9, +15.3]	0.0	0.0	0.0	0.5	2.7	0.596
P2 symmetric smoothing	46.2	+14.8 [+12.4, +17.0]	+14.5 [+11.4, +16.9]	+14.6 [+11.3, +17.6]	+7.6 [+5.4, +9.3]	+9.1 [+7.7, +10.6]	2.4	5.0	6.1	4.5	3.6	0.767
seed 2	42.0	+19.0 [+16.7, +21.5]	+20.6 [+17.3, +23.6]	+17.5 [+14.7, +20.5]	+9.0 [+7.2, +10.6]	+12.7 [+11.0, +14.3]	2.2	4.2	5.7	4.5	3.0	0.796
Effort scheme, no beautified views	39.7	+19.0 [+16.2, +21.7]	-11.3 [-15.8, -6.3]	-32.2 [-36.2, -27.5]	+11.2 [+8.4, +13.5]	-2.1 [-3.9, +0.1]	3.6	35.4	75.1	4.8	7.6	0.719
Effort scheme, symmetric smoothing	41.7	+22.4 [+20.0, +25.1]	+25.0 [+21.3, +28.5]	+24.9 [+19.3, +28.5]	+5.8 [+3.4, +7.9]	+7.2 [+5.8, +8.7]	2.0	4.2	6.1	7.3	4.3	0.757
released Effort fine-tuned, no beautified views	42.9	+17.2 [+12.3, +20.0]	-21.6 [-26.0, -14.6]	-39.9 [-44.7, -34.7]	+9.8 [+6.3, +12.2]	+1.1 [-3.2, +4.3]	4.2	39.2	81.2	7.7	6.4	0.754
released Effort fine-tuned, symmetric smoothing	43.7	+23.0 [+18.3, +26.4]	+6.6 [-1.3, +13.5]	-19.0 [-26.4, -10.1]	+10.0 [+5.9, +12.6]	-1.2 [-6.6, +2.9]	2.2	15.7	43.0	7.0	7.3	0.740
OURS, three-way, full data	47.0	-25.1 [-28.6, -21.8]	-42.3 [-46.3, -36.6]	-44.3 [-49.1, -38.3]	+4.5 [+1.7, +6.9]	+5.4 [+3.7, +6.6]	34.9	79.8	90.1	8.1	5.2	0.688
OURS, symmetric-view variant	43.5	+28.1 [+25.2, +30.6]	+33.8 [+29.9, +36.8]	+34.1 [+30.2, +37.1]	+12.8 [+11.0, +15.1]	+3.8 [+2.2, +5.1]	1.0	1.6	1.7	2.4	5.3	0.796
OURS, binary control	36.5	+0.7 [-3.2, +7.5]	-15.4 [-21.1, -2.5]	-15.0 [-20.8, -0.7]	+11.9 [+9.9, +14.0]	+6.2 [+5.1, +8.0]	9.9	35.5	45.6	4.7	5.4	0.754

The no-exposure arm reproduces the published detectors. P0, trained on FF++ with no beautified view, accuses 54.2, 56.1 and 43.5% (three seeds) of the Smoothing-90 renders at the 5%-FPR threshold of the service's originals, scores the render above its own original in 97.0–99.0% of pairs, and moves the median $P(\text{fake})$ from 0.05 to 0.97 (seed 1). Thirteen of the seventeen published detectors lie at 36.3–68.7% and fifteen at 89–99% under the same two readings. P0 is at the 5% floor on the other nine operation-by-level cells of the service (whitening 4.5–7.5%, face slimming 4.5–6.1%, eye enlarging 4.8–5.9%) and dose-dependent on smoothing (12.6–17.0, 29.5–40.8 and 43.5–56.1% at levels 30, 60 and 90). P0 trains with label-symmetric Gaussian blur, down-up scaling and JPEG on half of its triplets, so plain blur or re-compression in training does not prevent the accusation. With beautified views on genuine faces only (P1) the Smoothing-90 rate falls to 3.7, 7.1 and 4.6%; with views on both classes (P2) to 11.7, 7.1 and 6.9% (paired differences against P0 of -37 to -51

points, every interval below zero). The rate holds under every threshold rule tried (1% and 10% FPR, a fixed 0.5, thresholds set on Celeb-DF-v2 or DFD genuine frames). P2 still scores the render above its original in 72–84% of pairs (AUROC 0.63–0.70), so for P2 the accusation is lowered, not removed. Both arms pay for it on untreated Celeb-DF-v2: frame AUC -0.012 to -0.021 for P2 and -0.025 to -0.030 for P1 against P0 on every seed, intervals below zero, and 5–8 points of miss rate at 5% FPR. This first analysis was post hoc. Its per-epoch curve for the both-class arm (27% at epoch 2, 9–12% from epoch 5, with checkpoints chosen on FF++ validation AUC by margins of 0.0003–0.002) is disclosed for that reason; the arms below were declared before training. Inside Effort's SVD-residual scheme the P2 views leave 30.0 and 38.9% (P0 under that scheme 56.9 and 63.2%), and two epochs of fine-tuning the released Effort checkpoint with them leave 52.7% against 55.3% without them.

Necessary: views without smoothing do nothing. Declared reading: if the arm stays at 30% or more on both seeds, the effect is exposure to smoothing itself. The arm removes smoothing from every beautified view of P2 and keeps colour grades, the fourteen colour filters, whitening, eye and jaw edits, and the shared degradation. Smoothing-90 is then accused at 55.3% [51.9, 58.6] and 67.1% (two seeds), against P0 of the same seed $+1.2$ [-2.2, 4.0] and $+11.1$ [7.0, 14.8], and the render is scored above its original in 98.6 and 99.4% of pairs. The genuine-only version without smoothing gives 55.9% ($+1.8$ [-1.8, 5.2] against P0). The declared reading holds. These arms still lose untreated Celeb-DF-v2 video AUC against P0 (0.913 and 0.924 against 0.938 and 0.938), so that cost is not caused by smoothing alone.

Sufficient, but not free. The second arm makes smoothing the only beautified view, on both classes. Bars, against P0 of the same seed, on both seeds: (a) Smoothing-90 at most 15% at the 5%-FPR threshold of the originals; (b) untreated Celeb-DF-v2 and DFD video AUC not lower by more than 0.010; (c) Celeb-DF-B AUC under the presets not lower by more than 0.010. Smoothing-90 falls to 2.3% [1.7, 2.9] and 5.3% [4.4, 7.0], -51.9 [-56.1, -48.3] and -50.7 [-55.3, -46.5] against P0: (a) is met on both seeds. Bar (b) is not met as written: untreated Celeb-DF-v2 video AUC is 0.904 and 0.926 against 0.938 and 0.938, untreated DFD 0.967 and 0.966 against 0.981 and 0.979. Bar (c) is 0.865 and 0.870 against 0.860 and 0.883; the second seed is 0.013 lower and the bar is not met as written. The declared reading for "(a) only" is "sufficient but not free": every kind of beautified view we tried, with or without smoothing, costs 0.01–0.03 of untreated Celeb-DF-v2 video AUC (P2 0.917 and 0.920, no smoothing 0.913 and 0.924, smoothing only 0.904 and 0.926, P1 0.900 and 0.910).

The genuine-only recipe is an evasion route. Smoothing shown to genuine faces only lowers the accusation to 0.7, 1.0 and 0.9% (three seeds), scores the render above its original in 2.6, 6.5 and 5.4% of pairs (AUROC 0.095, seed 1), and has the best Celeb-DF-B numbers of the LoRA arms (AUC under presets 0.887, 0.896 and 0.893 against P0 0.860, 0.883 and 0.883). A check written before any smoothed forgery was scored asks what such a model does to a forgery that a forger smooths (Table S9). The genuine-only arm misses every smoothed forgery: miss rises of $+56$ to $+63$ points on the bilateral forms, $+38$ to $+40$ on the median filter and $+11$ to $+14$ on the Gaussian blend, with lower interval bounds of $+49.9$ or more on the bilateral forms, the forged score falling in 100% of frames and false alarms on smoothed genuine frames at 0.0%. P0 of the third seed rises by $+24.0$, $+14.3$, $+4.3$, $+12.6$ and $+5.6$. The arm with all beautified views on genuine faces only behaves the same ($+46$ to $+51$ on the bilateral forms). The shortcut "smoothed means genuine" is what removed the accusation, and it carries over to smoothing forms the views did not contain. The check fails on all three seeds, and the recipe is not recommended.

The both-class arms carry a shortcut of the opposite sign. A whole-crop bilateral filter on a *genuine* frame is called forged by the smoothing-only both-class arm in 47.8, 82.3 and 88.3% of frames at the three strengths, by the main model in 34.9, 79.8 and 90.1%, by P2 in 3.0, 31.8 and 49.1%, against P0 at 0.8, 16.8 and 39.6% and the binary control at 9.9, 35.5 and 45.6%. The cause is in the training code: genuine items receive the face-local scripted smoothing and pre-rendered smoothing frames, while forged items without landmarks receive only the whole-crop bilateral filter, so the form of the smoothing predicts the class and the model learned "whole-crop bilateral means forged". This is a class-asymmetric augmentation, and it affects every both-class arm and the main model.

A class-symmetric view. A smoothed view is drawn the same way wherever landmarks exist: with probability 0.5 the face-local scripted smoothing, otherwise the whole-crop bilateral filter at one of three strengths; forged frames without landmarks receive the whole-crop form; no pre-rendered smoothing frames. Both forms occur in both classes, in proportions that are not identical (genuine 0.5 / 0.5; forged with landmarks 0.33 / 0.67; forged without landmarks 0 / 1). Bars: (a) Smoothing-90 at most 15% at the 5%-FPR rule and the render scored above its original in between 25 and 75% of pairs; (b) false alarms on smoothed genuine Celeb-DF-v2 frames at most 15% for every form; (c) a miss rise on smoothed forgeries not more than 5 points above P0's for every form; two seeds. Bars (a) and (b) are met on both seeds: Smoothing-90 5.9 and 6.1%, the render above its original in 53.7 and 51.8% of pairs, false alarms on smoothed genuine frames 2.4, 5.0, 6.1, 4.5 and 3.6% (seed 1) and 2.2, 4.2, 5.7, 4.5 and 3.0% (seed 2) across the five forms. Bar (c) is not met as written: the miss rises are $+14.8$, $+14.5$, $+14.6$, $+7.6$ and $+9.1$ (seed 1) and $+19.0$, $+20.6$, $+17.5$, $+9.0$ and $+12.7$ (seed 2), against P0's $+27.0$, $+5.1$, -16.0, $+19.2$ and $+3.0$ and $+26.7$, $+6.1$, -11.8, $+18.0$ and $+8.1$, so the arm exceeds the reference

by more than 5 points at the mid and strong bilateral strengths on both seeds and under the Gaussian blend on the first (by 4.6 on the second). The reference itself is the reason: P0's miss rate *falls* under strong smoothing because P0 accuses everything smoothed, with false alarms of 39.6 and 42.9% on smoothed genuine frames at that strength, and the bar did not anticipate a reference that passes forgeries by accusing all smoothed faces. We keep the bar as written and report it as not met. Read without a threshold, with both classes smoothed, the AUC of the symmetric arm is 0.835, 0.796, 0.767, 0.829 and 0.850 (seed 1) and 0.849, 0.809, 0.796, 0.853 and 0.865 (seed 2) against P0's 0.848, 0.776, 0.747, 0.852 and 0.868 and 0.852, 0.781, 0.736, 0.855 and 0.871: above P0 at the two heavier bilateral strengths, lower by at most 0.023 elsewhere. On untreated data its Celeb-DF-v2 video AUC is 0.919 and 0.934 against P0's 0.938 and 0.938 and its DFD video AUC 0.972 and 0.968 against 0.981 and 0.979. On Celeb-DF-B its AUC under the presets is 0.866 and 0.888 against 0.860 and 0.883, its beautification-specific miss rise $+14.2$ and $+10.4$ against $+17.0$ and $+14.2$, and its false-alarm rate on beautified genuine frames 5.5% (seed 1). On DF40 its video AUC is 0.968 and 0.950 on face swap and 0.871 and 0.796 on reenactment in the FF++ domain, 0.913 and 0.935 and 0.670 and 0.646 in the Celeb-DF domain (Table S10).

Under Effort's tuning scheme. Declared: the same view, FF++ only, in Effort's scheme from CLIP and as a two-epoch fine-tune of the released Effort checkpoint (learning rate 2×10^{-5}), one seed, bars (a)–(c) against each scheme's own reference; both pass: the recipe is reported as scheme-independent; one passes: reported per scheme; neither: the LoRA result stands alone. Effort scheme from CLIP against its reference: (b) false alarms on smoothed genuine frames 2.0, 4.2, 6.1, 7.3 and 4.3% (reference 3.6, 35.4, 75.1, 4.8 and 7.6%), met. (a) Smoothing-90 14.5% at the 5%-FPR rule (reference 56.9%), but the render above its own original in 88.4% of pairs (band 25–75%), not met. (c) miss rise $+22.4$, $+25.0$, $+24.9$, $+5.8$ and $+7.2$ against $+19.0$, -11.3, -32.2, $+11.2$ and -2.1, not met as written, with the same cause as above. Released Effort fine-tuned for two epochs: Smoothing-90 39.9%, above own original 96.6% (reference 55.3% and 98.5%), (a) not met. Neither passes, so the LoRA result stands alone. The view removes the accusation of scripted smoothing under both schemes and of the commercial service only under LoRA; under the Effort scheme the commercial renders are still ranked above their originals. The recipe is not scheme-independent.

The main model with the symmetric view. Declared: the recipe of the main model with the symmetric view (the scripted smoothing of a genuine view is, with probability 0.5, the whole-crop bilateral filter; pre-rendered smoothing frames dropped; everything else unchanged), seed 20260929, checkpoint by the main model's rule. It replaces the main model if, against the current main model, (i) whole-crop-smoothed genuine Celeb-DF-v2 false alarms are at most 15% at every strength, (ii) Smoothing-90 is at most 15% with the render above its original in 25–75% of pairs, (iii) untreated Celeb-DF-v2 and DFD video AUC are not lower by more than 0.010, (iv) the explanation flip rate on the three part-forgery editors is not lower by more than 1.0 point, (v) Celeb-DF-B AUC under the presets is not lower by more than 0.010; otherwise the current main model stays and the asymmetry is stated as a limitation. Result: (i) 1.0, 1.6 and 1.7% (was 34.9, 79.8 and 90.1%), met. (ii) 8.1% and 30.0% (was 8.6% and 32.5%), met. (iii) Celeb-DF-v2 0.954 (0.947) and DFD 0.983 (0.978), met, both higher. (iv) flip rate on donor transplant, SD-1.5 and SDXL inpainting 92.6, 92.2 and 91.9% (94.0, 93.6 and 93.5%), lower by 1.4, 1.4 and 1.6 points: not met as written. The ceiling fell by the same amount (92.6, 92.1 and 92.0%) and IoU_7 is unchanged (0.836, 0.843 and 0.833). (v) Celeb-DF-B AUC under the presets 0.868 (0.866), with a beautification-specific miss rise of $+10.0$ ($+6.7$) and false alarms on beautified genuine frames of 9.2% (9.5%). Also: validation selection score 0.871 (0.862); untouched originals of the unseen service sent to the beautified class 13.9% (20.8%); frame AUC with both classes smoothed 0.878, 0.831, 0.796, 0.880 and 0.878 (0.803, 0.740, 0.688, 0.819 and 0.855); miss rise on smoothed forgeries $+28.1$, $+33.8$, $+34.1$, $+12.8$ and $+3.8$, where the old model's -25 to -44 came from accusing everything smoothed. By the rule, (iv) fails, so the current main model stays and the retrained model is reported next to it as the symmetric-view variant. On DF40 the variant reaches 0.952 and 0.909 on face swap and 0.738 and 0.539 on reenactment (main model 0.952, 0.905, 0.718 and 0.502), so it remains behind the CLIP detectors on reenactment.

S10 DF40: every detector and arm

Table S10: DF40 held-out, untreated. Video AUC (mean frame score per video), mean over methods, with 95% bootstrap intervals over videos. Real videos are the FF++ and Celeb-DF-v2 test reals; the 718 MobileSwap videos from FF++ training identities were removed; 140 fake videos per method in the Celeb-DF domain. FS: nine face-swap methods; FR: twelve reenactment methods. CADDM and the rows from P0 downwards were scored after the first table had been read.

	FF++ domain		Celeb-DF domain	
Detector	FS	FR	FS	FR
Xception	0.854 [0.839, 0.869]	0.859 [0.846, 0.872]	0.670 [0.653, 0.688]	0.691 [0.674, 0.706]
EfficientNet-B4	0.812 [0.796, 0.828]	0.818 [0.804, 0.831]	0.724 [0.706, 0.743]	0.712 [0.697, 0.729]
F3Net	0.826 [0.809, 0.841]	0.797 [0.782, 0.811]	0.697 [0.679, 0.715]	0.671 [0.654, 0.686]
SPSL	0.766 [0.748, 0.783]	0.754 [0.737, 0.770]	0.717 [0.699, 0.739]	0.704 [0.688, 0.721]
UCF	0.843 [0.828, 0.857]	0.825 [0.811, 0.839]	0.716 [0.698, 0.733]	0.704 [0.689, 0.720]
RECCE	0.860 [0.846, 0.874]	0.850 [0.837, 0.863]	0.736 [0.719, 0.753]	0.714 [0.698, 0.729]
CORE	0.850 [0.835, 0.865]	0.858 [0.846, 0.870]	0.656 [0.638, 0.674]	0.676 [0.659, 0.693]
SRM	0.836 [0.820, 0.852]	0.830 [0.816, 0.842]	0.732 [0.715, 0.750]	0.740 [0.725, 0.756]
Effort [1]	0.944 [0.935, 0.953]	0.808 [0.793, 0.822]	0.935 [0.926, 0.944]	0.710 [0.692, 0.726]
Forensics Adapter [2]	0.948 [0.940, 0.956]	0.828 [0.814, 0.841]	0.897 [0.885, 0.908]	0.665 [0.649, 0.681]
GenD [3]	0.966 [0.959, 0.973]	0.935 [0.927, 0.943]	0.947 [0.940, 0.955]	0.828 [0.815, 0.840]
DFD-HR [4]	0.982 [0.977, 0.986]	0.932 [0.924, 0.940]	0.940 [0.931, 0.948]	0.698 [0.681, 0.714]
CADDM [6]	0.869 [0.856, 0.882]	0.794 [0.779, 0.809]	0.788 [0.772, 0.804]	0.541 [0.523, 0.560]
OURS, three-way, full data	0.952 [0.943, 0.960]	0.718 [0.701, 0.733]	0.905 [0.894, 0.915]	0.502 [0.484, 0.519]
second seed	0.960 [0.952, 0.967]	0.780 [0.764, 0.793]	0.928 [0.918, 0.937]	0.559 [0.540, 0.577]
OURS, symmetric-view variant	0.952 [0.944, 0.959]	0.738 [0.722, 0.752]	0.909 [0.898, 0.919]	0.539 [0.521, 0.556]
OURS, binary control	0.969 [0.962, 0.975]	0.842 [0.828, 0.854]	0.926 [0.916, 0.936]	0.647 [0.630, 0.663]
OURS, FF++ only, three-way	0.962 [0.954, 0.969]	0.809 [0.794, 0.823]	0.933 [0.923, 0.942]	0.637 [0.621, 0.653]
P0	0.981 [0.976, 0.985]	0.906 [0.896, 0.915]	0.925 [0.916, 0.935]	0.692 [0.675, 0.707]
P0, seed 2	0.970 [0.964, 0.975]	0.877 [0.865, 0.887]	0.939 [0.930, 0.948]	0.720 [0.704, 0.734]
P0, seed 3	0.979 [0.974, 0.983]	0.873 [0.861, 0.884]	0.931 [0.923, 0.940]	0.633 [0.616, 0.648]
P2	0.942 [0.933, 0.950]	0.809 [0.794, 0.823]	0.907 [0.895, 0.917]	0.644 [0.627, 0.659]
P2, seed 2	0.944 [0.935, 0.952]	0.785 [0.770, 0.799]	0.901 [0.890, 0.912]	0.643 [0.627, 0.658]
P2 no smoothing	0.953 [0.945, 0.960]	0.858 [0.846, 0.870]	0.883 [0.870, 0.894]	0.705 [0.689, 0.720]
P2 smoothing only	0.945 [0.935, 0.953]	0.779 [0.762, 0.794]	0.908 [0.897, 0.919]	0.593 [0.575, 0.610]
P2 smoothing only, seed 2	0.953 [0.945, 0.960]	0.833 [0.818, 0.847]	0.921 [0.911, 0.931]	0.629 [0.611, 0.645]
P1	0.952 [0.944, 0.960]	0.807 [0.792, 0.821]	0.885 [0.874, 0.897]	0.569 [0.551, 0.587]
P1 smoothing only	0.965 [0.958, 0.971]	0.871 [0.859, 0.882]	0.925 [0.916, 0.934]	0.673 [0.656, 0.689]
P1 smoothing only, seed 2	0.963 [0.956, 0.969]	0.869 [0.857, 0.880]	0.940 [0.932, 0.948]	0.689 [0.673, 0.705]
P1 smoothing only, seed 3	0.967 [0.960, 0.973]	0.900 [0.889, 0.910]	0.923 [0.913, 0.932]	0.679 [0.663, 0.695]
P2 symmetric smoothing	0.968 [0.961, 0.974]	0.871 [0.858, 0.882]	0.913 [0.902, 0.923]	0.670 [0.653, 0.686]
P2 symmetric smoothing, seed 2	0.950 [0.941, 0.958]	0.796 [0.780, 0.810]	0.935 [0.926, 0.944]	0.646 [0.628, 0.661]
Effort scheme, symmetric smoothing	0.966 [0.960, 0.973]	0.822 [0.808, 0.835]	0.934 [0.925, 0.943]	0.614 [0.596, 0.631]
released Effort fine-tuned, symmetric smoothing	0.941 [0.931, 0.951]	0.805 [0.790, 0.819]	0.920 [0.910, 0.930]	0.705 [0.688, 0.722]
paired difference, OURS (three-way, full data) minus				
Effort [1]	+0.007 [-0.002, +0.018]	-0.090 [-0.109, -0.072]	-0.030 [-0.043, -0.018]	-0.208 [-0.228, -0.187]
Forensics Adapter [2]	+0.004 [-0.006, +0.012]	-0.110 [-0.126, -0.094]	+0.009 [-0.006, +0.021]	-0.163 [-0.184, -0.143]
GenD [3]	-0.015 [-0.023, -0.007]	-0.218 [-0.233, -0.203]	-0.042 [-0.053, -0.032]	-0.326 [-0.343, -0.308]
DFD-HR [4]	-0.031 [-0.038, -0.024]	-0.214 [-0.229, -0.199]	-0.035 [-0.046, -0.024]	-0.196 [-0.213, -0.178]
binary control	-0.017 [-0.023, -0.012]	-0.124 [-0.135, -0.113]	-0.021 [-0.029, -0.013]	-0.145 [-0.157, -0.133]

The second seed of the main model is 0.780 and 0.559 on reenactment. On four Celeb-DF-domain reenactment methods (TPSM, DaNet, MCNet, FaceVid2Vid) the main model scores 0.27–0.37. Across our arms, reenactment in the FF++ domain is P0 0.906 (second and third seed 0.877 and 0.873), P2 without smoothing 0.858, P2 0.809 (0.785), P2 with smoothing only 0.779 (0.833), P1 0.807, the three-way FF++-only model 0.809, the binary full-data control 0.842 and the main model 0.718; in the Celeb-DF domain 0.692 (0.720 and 0.633), 0.705, 0.644 (0.643), 0.593 (0.629), 0.569, 0.637, 0.647 and 0.502. The three seeds of P0 span 0.09 in the Celeb-DF domain, so differences between arms smaller than that are within the seed spread of the reference there. Four caveats attach to the table. The real videos are the FF++ and Celeb-DF-v2 test reals already used elsewhere in the paper, and Celeb-DF has no identity-disjoint split. Every detector in the table, ours included, is trained on FF++. Our models were trained on a different crop from the released 256 px DF40 crops; Table S2 gives the size of that handicap on Celeb-DF-v2 and DFD. SBI and LAA-Net are absent because their pipelines need their own crops from full frames. DF40's face-editing set (five GAN and diffusion attribute editors on CelebA) contains no beautification or retouching, so DF40 does not test the beautified class.

S11 Fusion of a texture-based detector with CADDM

Table S11: Score fusion with released weights and no training. Each detector's score is replaced by its quantile among the genuine scores of the same source (for the retouching service: its untouched originals); the fused score is the mean, the minimum or the product of the two quantiles. Columns as in the paper: commercial Smoothing-90 above the 5%-FPR threshold and above its own original, untreated video AUC, DF40 video AUC.

	commercial Smoothing-90		untreated video AUC		DF40, FF++ domain		DF40, Celeb-DF domain	
Score	above 5%-FPR thr. (%)	above own orig. (%)	Celeb-DF-v2	DFD	FS	FR	FS	FR
GenD [3] with CADDM [6]								
first member alone	68.7	98.5	0.954	0.977	0.966	0.935	0.947	0.828
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	28.7	92.2	0.970	0.964	0.955	0.942	0.914	0.761
fusion, minimum	23.4	78.2	0.962	0.956	0.925	0.922	0.867	0.708
fusion, product	27.6	87.2	0.969	0.966	0.944	0.935	0.895	0.733
DFD-HR [4] with CADDM [6]								
first member alone	64.9	99.0	0.943	0.976	0.982	0.932	0.940	0.698
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	25.0	91.5	0.963	0.964	0.964	0.942	0.912	0.673
fusion, minimum	21.8	78.1	0.956	0.955	0.932	0.924	0.866	0.659
fusion, product	24.5	87.2	0.962	0.967	0.951	0.935	0.893	0.658
Effort [1] with CADDM [6]								
first member alone	43.2	95.6	0.930	0.972	0.944	0.808	0.935	0.710
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.6	86.9	0.965	0.966	0.945	0.881	0.910	0.686
fusion, minimum	22.0	77.4	0.956	0.957	0.915	0.853	0.868	0.658
fusion, product	23.7	84.3	0.963	0.967	0.935	0.869	0.894	0.667
OURS P0 (FF++ only, no beautified views) with CADDM [6]								
first member alone	54.2	98.5	0.938	0.981	0.981	0.906	0.925	0.692
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.4	91.9	0.961	0.969	0.960	0.939	0.893	0.678
fusion, minimum	20.0	79.1	0.955	0.962	0.929	0.923	0.850	0.665
fusion, product	24.2	87.6	0.961	0.971	0.949	0.933	0.876	0.663

Declared before any fused score was computed: the pairs (GenD, DFD-HR, Effort and our P0, each with CADDM), the three rules, and the reading. A fusion counts as useful only if, for the same pair and rule, Smoothing-90 is at most half of the first member's rate, reenactment video AUC is within 0.03 of the first member in both domains, and untreated video AUC is not below the better member by more than 0.01; one rule meeting this for at least two of the three published pairs counts, and anything else is reported as moving the operating point along the same trade-off. No rule meets all three conditions for two published pairs (Table S11). Under the mean rule GenD fails the reenactment condition in the Celeb-DF domain (-0.067); DFD-HR and our P0 fail the untreated condition on DFD by 0.002 (-0.012 against 0.010); Effort fails the halving condition by 3 points. The minimum rule cuts the accusation further (20.0–23.4%; product 23.7–27.6%), and both lose more reenactment in the Celeb-DF domain (0.66–0.73). Descriptive: the accusation of smoothed genuine faces falls by 43–62% without training; untreated Celeb-DF-v2 AUC of the mean fusion exceeds both members in all four pairs ($+0.016$ to $+0.035$ over the better member); FF++-domain reenactment is kept or improved in all four; the loss is in Celeb-DF-domain reenactment, where CADDM is at 0.54, and on DFD, where CADDM is 0.06 below the other members. Renders are still ranked above their own originals in 87–92% of pairs under the mean rule: the fusion lowers the rate at a threshold, not the ordering.

S12 Methods that did not meet their bars

Table S12: Every method arm tried under a pre-registered bar. FF++ only unless noted; each block starts with its reference. Bars for the FF++-only arms: AUC_b on Celeb-DF-B ≥ 0.890 and $\Delta\text{MISS}_{\text{beauty}} \leq 7.5$, untreated Celeb-DF-v2 AUC within 0.01 of the reference. None is met.

	Celeb-DF-B				Celeb-DF-v2	DFD filters
Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	AUC untreated	AUC_f
random colour grading (reference)	0.938	0.872	+7.5	7.8	0.898	0.932
no grading	0.929	0.849	+14.1	7.2	0.880	0.917
operator, random edits	0.932	0.876	+7.9	7.7	0.873	0.932
operator, worst of 4 draws	0.928	0.867	+12.9	5.6	0.860	0.935
operator, 1-step search	0.930	0.864	+8.2	8.3	0.876	0.935
operator, search within measured edits	0.928	0.869	+15.1	5.2	0.874	0.934
operator, consistency objective	0.924	0.856	+13.0	5.7	0.849	0.922
operator, 3-step search	0.922	0.859	+8.9	7.6	0.849	0.937
decision pooled from the evidence map	0.937	0.877	+14.9	5.0	0.889	0.924
same, no beautified views	0.942	0.891	+14.7	4.4	0.899	0.932
full data, binary (reference)	0.949	0.896	+7.0	7.4	0.914	0.952
full data, binary, operator search	0.942	0.892	+9.7	6.0	0.911	0.945
released Effort, mapped (reference)	0.909	0.808	+11.2	4.7	0.886	0.904
released Effort + 2 epochs, no beautified views	0.903	0.805	+5.8	9.4	0.866	0.918
released Effort + 2 epochs, both classes	0.895	0.803	+2.7	10.3	0.861	0.916

The operator. A differentiable edit $B(x; u)$ on a face crop in $[0, 1]$ with every component bounded by a sigmoid: per-channel gamma, gain and lift; a 3×3 colour matrix close to the identity; saturation from monochrome to 1.7; contrast about the mean; a 4×4 luminance field and a per-channel tint field, upsampled bilinearly; a blend toward a Gaussian-smoothed copy; additive grain. Its parameters were fitted by gradient descent to 1,000 pairs each of four measured families (Table S13); the search of the "measured edits" arms starts from a fitted edit and is projected onto the 0.5th–99.5th percentile range of the fitted parameters.

Table S13: The operator fitted to measured edits (training split only). The residual of the commercial renders is geometry, which a photometric operator cannot express.

Measured edit	n	PSNR before	PSNR after fit	10th pct
Tencent retouching	1000	21.4	28.6	24.8
Megvii retouching	1000	29.1	30.6	28.0
training colour filters	1000	21.0	40.1	32.0
random grading	1000	24.2	48.4	39.0

The search works; the training does not carry over. On 400 FF++ validation frames the reference model keeps 96.0% of genuine faces genuine without an edit, 71.0% under a random draw of the operator, 41.0% under the worst of four draws, 8.5% under a one-step signed-gradient search and 0.0% under ten steps; forged faces stay forged in 93.5, 89.5, 83.0, 74.0 and 20.0% of cases. After training on the one-step worst case the genuine accuracy under that search is 71.5% (25.0% under ten steps), and after the three-step variant 76.5% (41.5%). The gain does not carry over to the app presets or to filters from a library the operator was not fitted to, and the forged accuracy under random edits falls from 89.5 to 79.5%: with a label loss, edited views of genuine faces dominate the signal and the model learns that a heavily edited face is less likely forged.

A learned inverse. The exact algebraic inverse of the invertible part of the operator restores an edited frame to 53–58 dB when the true parameters are given. A ResNet-18 estimator trained to predict them from the edited frame alone reaches 26.7 dB on random edits (25.9 before) and 22.7 dB on the held-out filters (21.0 before), and moves untouched frames to 34.4 dB. In front of the detector it changes the false alarms under the held-out filters from 21.5 to 23.6%. A single image does not determine its colour transform.

Edit sensitivity as evidence. The response of the forged logit to six fixed probe edits is larger for genuine than for forged frames on average, but inside the deciles of the clean logit where both classes occur the two respond alike (1.15 vs 0.96; 0.48 vs 0.10), so the asymmetry is a ceiling effect; adding the responses to the logit leaves the false alarms under filters at 16–22%.

A decision pooled from the evidence map. With a zero map target on every beautified item and the forged logit taken as the mean of the top 5% map cells, the map stays dark on filtered genuine faces (mean logit -10.2 against $+1.9$ for the edit map) and the false alarms under filters fall from 21.5 to 14.7%, but the beautification-specific miss rise on Celeb-DF-B grows from 7.5 to 14.9 points.

Fine-tuning a released checkpoint. The released Effort weights were loaded into our pipeline (the mapped model reproduces the released scorer: Celeb-DF-B AUC 0.909, AUC_b 0.808) and fine-tuned for two epochs. With both-class exposure the miss rise falls from 11.2 to 2.7 points, the false alarms on beautified genuine faces rise from 4.7 to 10.3%, AUC_b stays at 0.803 and the untreated AUC falls by 0.014; without beautified views the same fine-tuning gives 5.8 points and 9.4%. The operating point moves; separability does not.

S13 Commercial retouching: the native three-way reading and all variants

The paper reports the retouching service under its one rule (score $P(\text{fake})$, threshold at 5% FPR on the service's originals) and without a threshold. Table S14 gives the other reading, the native three-way output of the models that have a beautified class; its numbers are not comparable with the one-rule numbers.

Table S14: Retouching by a service never seen in training, native three-way rule: 2,210 FFHQ originals and their renders (four operations at three levels, 26,519 images). Paired rank: share of pairs in which the render gets a higher $P(\text{filter})$ than its own original (50 means no signal).

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Trained with two vendors' renders and their single-operation components; Alibaba never seen (ours)							
OURS (CLIP ViT-L/14 + LoRA)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
OURS-lite (EfficientNet-B4)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4

Three-way models trained on FF++ alone separate retouched from untouched faces at chance and call commercial smoothing fake, up to 86% of faces at the strongest level. Trained with the single-operation components of two other services, OURS calls 0.6% of the retouched faces fake and 0.8% at the strongest smoothing by its own three-way decision (under the one rule: 4.7% and 8.6%), sends 67.7% of them to FILTER (smoothing at level 90: 98.8%) and gives 92.9% of renders a higher filter score than their own original. It also sends 20.8% of the untouched originals to FILTER, so its balanced accuracy on "retouched or not" is 73.5% (75.4% for the second seed), against a pre-registered target of 75%, not met as written by the first seed. The AUROC of $P(\text{filter})$ between originals and renders is 0.80–0.82 for every model trained with components, so no threshold reaches the target with margin: a single light operation moves the score by about one standard deviation of the spread between photographs. The symmetric-view variant sends 13.9% of the untouched originals to FILTER and has a balanced accuracy of 73.4%.

Under the one rule the accusation concentrates on smoothing. Table S15 gives the share of renders above the 5%-FPR threshold of the originals per operation at the strongest level and for the three smoothing levels. The other three operations stay at 3.7–8.1% for thirteen of the sixteen usable detectors; Effort reaches 16.8% on eye enlarging, UnivFD 11.7% and DFD-HR 9.6%. Smoothing is dose-dependent for every detector except CADDM.

Table S15: The retouching service per operation under the one rule: share of renders (%) above the threshold that accuses 5% of the untouched originals. Level 90 for the four operations, and the three levels of smoothing.

	level 90			smoothing		
Detector	whitening	face slimming	eye enlarging	level 30	level 60	level 90
Xception	5.7	3.7	4.6	16.6	36.2	51.6
EfficientNet-B4	5.1	4.0	4.7	11.5	23.9	36.3
F3Net	5.4	4.2	4.4	15.0	33.2	50.6
SPSL	5.2	4.9	4.9	14.9	34.1	51.2
UCF	6.6	4.0	4.2	16.8	32.5	41.4
RECCE	5.3	4.1	4.5	11.9	22.7	36.8
CORE	5.7	4.4	4.8	14.8	36.6	54.8
SRM	4.5	3.8	4.7	18.7	45.0	62.8
SBI [5]	5.4	6.2	5.5	14.8	31.6	37.3
UnivFD	6.2	5.7	11.7	6.8	12.3	16.6
NPR	n/a	n/a	n/a	n/a	n/a	n/a
Effort [1]	11.1	9.6	16.8	13.2	27.2	43.2
Forensics Adapter [2]	5.4	6.2	7.0	10.2	25.8	39.7
GenD [3]	8.1	6.4	7.5	20.7	55.3	68.7
DFD-HR [4]	7.7	8.0	9.6	17.2	46.7	64.9
CADDM [6]	4.7	6.0	5.9	5.4	5.6	6.3
LAA-Net [7]	5.2	4.8	4.8	11.5	17.2	17.2
Ours, three-way, full data	4.8	3.7	3.6	6.2	5.0	8.6
Ours, binary control	5.4	5.0	4.3	5.6	6.3	9.1
Ours, symmetric-view variant	6.3	3.0	3.3	7.3	5.3	8.1
Ours, FF++ only, P0	6.8	5.8	5.4	17.0	40.8	54.2
Ours, FF++ only, P2 symmetric smoothing	5.8	5.0	5.0	4.5	3.9	5.9

Table S16: All trained variants on the unseen service (2,210 originals, 26,519 renders). Columns as in Table S14.

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, RepViT-M0.9	0.8/2.5/96.7	92.6	6.4	47.9	48.4	73 / 55 / 52 / 15	17.7
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, hybrid fakes (HYB)	33.8/16.7/49.5	54.6	22.1	52.6	69.2	92 / 83 / 65 / 44	50.2
HYB, faithful blend	28.2/22.1/49.6	52.8	28.0	51.6	65.7	92 / 77 / 62 / 35	58.4
HYB + symmetric degradation	31.7/26.3/42.0	45.3	33.6	51.7	64.3	89 / 76 / 63 / 35	63.1
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Vendor-trained variants (all single Alibaba reads)							
RU1 renders only (EffNet-B4, s1)	93.4/0.8/5.8	38.1	0.6	66.1	92.8	99 / 85 / 95 / 100	0.1
RU1 renders only (RepViT)	92.8/1.4/5.9	36.1	1.1	65.1	92.6	96 / 89 / 93 / 100	0.3
RU2 + components (EffNet-B4, s1)	76.8/0.5/22.8	68.2	0.3	72.7	94.2	99 / 90 / 94 / 100	0.4
RU2 + components (EffNet-B4, s2)	66.6/0.3/33.2	74.9	0.2	70.8	94.2	99 / 92 / 93 / 99	0.4
RU2 + components (RepViT)	43.4/0.9/55.7	83.7	0.5	64.0	95.1	99 / 91 / 96 / 99	1.1
RU3 cleaned components (EffNet-B4, s1)	76.7/0.4/22.9	68.8	0.3	72.9	94.9	99 / 92 / 94 / 99	0.5
RU3 + evidence head (EffNet-B4, s1)	83.2/0.6/16.2	62.2	0.5	73.0	94.1	99 / 91 / 93 / 99	0.8
RU4 + part edits (lite) (EffNet-B4, s1)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4
RU3 + evidence head, CLIP (CLIP-L/14, s1)	77.4/0.4/22.2	71.4	0.2	74.6	93.8	98 / 90 / 94 / 100	0.2
RU3 + evidence head, CLIP (CLIP-L/14, s2)	73.0/0.3/26.7	73.6	0.2	73.4	95.1	99 / 94 / 96 / 100	0.2
RU4 + part edits, CLIP (filter-F1 selection) (CLIP-L/14, s1)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100	0.0
RU4 + part edits, CLIP (CLIP-L/14, s1)	89.5/0.6/9.8	59.2	0.5	74.7	95.5	98 / 94 / 96 / 100	0.1
RU4 + part edits, CLIP (CLIP-L/14, s2)	83.3/0.4/16.2	67.5	0.2	75.6	93.0	98 / 88 / 95 / 100	0.2
RU5 without colour grading (CLIP-L/14, s1)	82.1/0.6/17.3	65.7	0.3	74.2	92.8	97 / 89 / 93 / 99	0.4
RU5 without vendor edit on forgeries (CLIP-L/14, s1)	83.8/0.4/15.8	65.6	0.2	74.9	94.7	98 / 94 / 94 / 100	0.1
RU5 data, three-way softmax (CLIP-L/14, s1)	81.5/0.7/17.8	66.4	0.5	74.3	94.1	98 / 94 / 93 / 99	0.5
RU5, CLIP (main) (CLIP-L/14, s1)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
RU5, CLIP (main) (CLIP-L/14, s2)	79.8/0.5/19.6	70.5	0.4	75.4	94.3	98 / 92 / 95 / 99	0.6

S14 Explanations: all variants

Table S17: Restoration test on held-out part-level forgeries for every variant. Columns as in the explanation table of the paper.

	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
Map	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
no head, Grad-CAM [8]	0.367	50.9	85.8	15.2	0.259	45.4	85.2	25.0	0.269	48.0	85.9	23.6
no head, Grad-CAM++ [9]	0.414	51.7	85.8	7.2	0.327	47.8	85.2	16.7	0.330	48.8	85.9	19.1
no head, Score-CAM [10]	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, part edits, FF++ only	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
+ symmetric degradation	0.723	86.3	87.1	0.4	0.719	82.5	83.7	2.0	0.728	83.7	84.4	1.5
+ restoration-consistency loss	0.759	89.4	90.0	0.4	0.614	75.2	79.8	4.4	0.684	82.5	84.6	2.6
+ consistency + symmetric degradation	0.722	88.4	89.3	0.7	0.723	85.2	86.1	2.4	0.732	86.1	86.9	1.8
head, vendor data, no part edits (EffNet-B4)	0.418	61.4	82.4	12.0	0.280	39.5	63.5	23.4	0.315	46.2	72.3	22.5
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
OURS-lite (EffNet-B4)	0.807	87.9	88.0	0.0	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
RU4, CLIP (filter-F1 selected, superseded)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5
OURS (CLIP ViT-L/14 + LoRA)	0.834	94.0	94.1	0.1	0.841	93.6	93.7	0.4	0.831	93.5	93.6	0.3

The donor-transplant test set uses donors from test videos only. An earlier version of that set drew donors from training videos (1,023 donor-frame and part pairs occurred in training); it was rebuilt, and the numbers of the main model moved by less than 0.2 points. The symmetric-view variant of the main model reaches flip rates of 92.6, 92.2 and 91.9% on the three editors against ceilings of 92.6, 92.1 and 92.0%, with IoU_7 of 0.836, 0.843 and 0.833. Figure S2 shows examples for the main model.

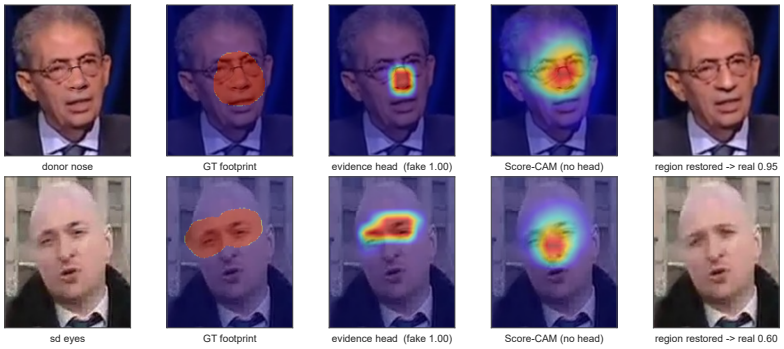


Figure S2: Held-out part-level forgeries: input, exact footprint, evidence map with the three-way decision, Score-CAM on the same backbone without the head, and the result of restoring the named region to the original.

S15 Sources that were examined and excluded

AR beauty filters. On paired Instagram AR-filter images, a model trained with such pairs called 94.8% of the filtered images *filter* with the face hull greyed out: the capture pipeline, not the face, carries the label. No claim is made on AR filters. **WildDeepfake.** The 800 test frames available to us come from six sequences, which gives intervals of 0.3–0.5 AUC; not reported. **DFDC, DFDCP.** Not available to us in a form with genuine and forged frames from the same pipeline. DF40 is evaluated (Table S10); only its face-swap and reenactment sets are used.

References

[1] Z. Yan, J. Wang, P. Jin, K.-Y. Zhang, C. Liu, S. Chen, T. Yao, S. Ding, B. Wu, and L. Yuan, "Orthogonal subspace decomposition for generalizable AI-generated image detection," in *ICML*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 70 268–70 288.

[2] X. Cui, Y. Li, A. Luo, J. Zhou, and J. Dong, "Forensics adapter: Adapting CLIP for generalizable face forgery detection," in *CVPR*, 2025, pp. 19 207–19 217.

[3] A. Yermakov, J. Cech, J. Matas, and M. Fritz, "Deepfake detection that generalizes across benchmarks," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

[4] J. Sun *et al.*, "DFD-HR: Generalizable deepfake detection via hierarchical routing learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[5] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *CVPR*, 2022.

[6] S. Dong, J. Wang, R. Ji, J. Liang, H. Fan, and Z. Ge, "Implicit identity driven deepfake face swapping detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 4490–4499.

[7] D. Nguyen, N. Mejri, I. P. Singh, P. Kuleshova, M. Astrid, A. Kacem, E. Ghorbel, and D. Aouada, "LAA-Net: Localized artifact attention network for quality-agnostic and generalizable deepfake detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 17 395–17 405.

[8] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[9] A. Chattopadhay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847.

[10] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, "Score-CAM: Score-weighted visual explanations for convolutional neural networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 24–25.